

# Deriving, Testing, and Bounding the Multi-Power Law for Schedule-Aware Loss-Curve Prediction

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Deep Learning — Task 2 | [github.com/wjjpku/DL-final](https://github.com/wjjpku/DL-final)

**One-line summary:** we reproduce the two empirical loss-curve laws (Tissue, MPL), *derive* MPL from SGD dynamics, confirm its predicted scale-invariance (all 27 curves collapse onto one master curve,  $R^2=0.997$ ), try two principled ways to *improve* it (honest result: MPL is near-optimal), and *localize* its one empirical parameter  $\gamma$  to the edge of stability.

# Outline

- ① Problem & Motivation
- ② Background & Reproduction
- ③ Deriving MPL from SGD Dynamics
- ④ Scale Invariance & a Universal Master Curve
- ⑤ Method Development: Can We Improve MPL?
- ⑥ The Edge-of-Stability Origin of  $\gamma$
- ⑦ Conclusion

# The problem: schedule-aware loss-curve prediction

**Goal.** Given a learning-rate (LR) schedule  $\{\eta_t\}$ , predict the whole pretraining loss curve  $L(t)$  *before* running the training.

## Why it matters.

- Pretraining LLMs is expensive; the LR schedule strongly shapes the entire loss trajectory, not just the endpoint.
- Classical scaling laws (Kaplan, Chinchilla) predict only the *final* loss vs. model size / data — they say nothing about the curve's response to the schedule *shape* (warmup, decay form, horizon).
- A good law (i) replaces trial runs with a formula, (ii) exposes mid-training dynamics (the extra drop from LR decay), and (iii) can be inverted to *search* for better schedules.

**The defining test** — cross-schedule generalization: **fit on cosine schedules, predict the unseen WSD family.**

# What this project does

We go beyond reproduction and study the governing law itself:

1. **Reproduce** the two representative laws (Tissue 2024; Multi-Power Law / MPL, Luo 2025) on the cosine→WSD task.
2. **Derive** MPL's functional form from SGD on a power-law curvature spectrum — giving each parameter a physical meaning and a prediction.
3. **Test** the prediction: kernel exponents are scale-invariant  $\Rightarrow$  all curves collapse onto one master curve.
4. **Improve?** Two principled methods (SC-MPL; a new  $(S, Q)$  asymptotic) — honest finding: MPL is near-optimal here.
5. **Bound:** localize MPL's one empirical parameter  $\gamma$  to the non-linear *edge of stability*.

Data: official public loss curves, 3 scales (25/100/400M)  $\times$  9 schedules.

# Background: the two empirical laws

Let  $S(t) = \sum_{i \leq t} \eta_i$  be the **cumulative LR** (a natural “time”), and  $\Delta\eta_k = \eta_{k-1} - \eta_k$  the LR decrement at step  $k$ .

**Tissue et al. (5 parameters)** — simple, interpretable, but coarse:

$$L(t) = L_0 + A S(t)^{-\alpha} - C S_2(t), \quad S_2 = \text{momentum-smoothed annealing area.}$$

**Multi-Power Law / MPL (7 parameters)** — current state of the art:

$$L(t) = L_0 + \underbrace{A S(t)^{-\alpha}}_{\text{backbone}} + \underbrace{B \sum_{k \leq t} \Delta\eta_k G(\eta_k^{-\gamma} (S(t) - S(k)))}_{\text{loss-drop term } LD}, \quad G(x) = 1 - (1 + Cx)^{-\beta}.$$

$LD$  models how *each* LR decrease keeps reducing the loss. The factor  $\eta_k^{-\gamma}$  is **fit empirically** in the original paper — we return to why.

# Reproduction: setup

## Setup (following the task).

- Data: official MPL loss curves, 25/100/400M, 9 schedules each.
- **Fit on cosine** schedules (cosine\_24000, cosine\_72000); **evaluate on 5 unseen WSD curves** (wsd, wsdld, wsdcon\_{3,9,18}).
- Objective: Huber loss on  $\log L$ ; optimizer: L-BFGS-B with multi-start.
- Engineering: Tissue's  $S_2$  rewritten as an exact linear recurrence ( $\sim 100\times$  faster); MPL's  $LD$  vectorized.

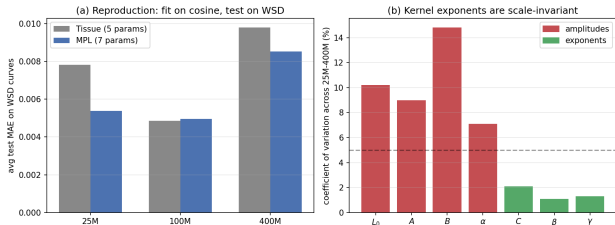
**Metric.** Average test MAE over the 5 WSD curves (lower is better).

Code: repro/reproduce\_cosine\_to\_wsd.py.

# Reproduction: results match the published numbers

| Scale | Tissue (5p)    | MPL (7p)       |
|-------|----------------|----------------|
| 25M   | 0.00783        | <b>0.00539</b> |
| 100M  | <b>0.00486</b> | 0.00496        |
| 400M  | 0.00979        | <b>0.00852</b> |

avg test MAE on WSD (fit on cosine)



**Analysis.** We recover the published behavior: MPL is more accurate at 25/400M and ties Tissue at 100M; MPL generalizes more stably across schedules. Our MPL test MAE (0.0054/0.0050/0.0085) matches Luo et al. **This well-fit MPL is the baseline the rest of the work must respect.**

# Why derive it? From 7 opaque parameters to physical meaning

**Model (assumptions).** Near the trajectory the loss is locally quadratic; in the Hessian eigenbasis, mode  $i$  has curvature  $\lambda_i$ , gradient-noise variance  $\sigma_i^2$ , and SGD is the linear noisy recurrence

$$\delta_i(t+1) = (1 - \eta_t \lambda_i) \delta_i(t) - \eta_t \xi_i(t), \quad L = L_{\min} + \frac{1}{2} \sum_i \lambda_i \mathbb{E}[\delta_i^2].$$

The second moment splits into a deterministic **bias** and a stochastic **noise floor**.

**(1) Bias  $\rightarrow$  power-law backbone.** With  $\delta^{\text{bias}} \approx \delta(0)e^{-\lambda S}$  and a power-law spectrum  $g(\lambda) \sim \lambda^a$ , a Tauberian argument gives

$$L_{\text{bias}}(S) = \int g(\lambda) e^{-2\lambda S} d\lambda \sim A S^{-\alpha}, \quad \alpha = a + 1, \quad L_0 = L_{\min}.$$



# The annealing term and a falsifiable prediction

**(2) Noise floor  $\rightarrow$  annealing term.** Solving the variance recurrence,  $V_i(t) \approx \sigma_i^2 \int_0^{S(t)} \eta(S') e^{-2\lambda_i(S(t)-S')} dS'$ ; at constant LR the noise loss is  $\propto \eta$  (**the floor tracks the current LR**). The bonus from decaying below the peak, written via the decrements, is

$$\Delta N(t) = \sum_k \Delta \eta_k \mathcal{K}(S(t) - S(k)) \quad \text{— exactly MPL's } LD.$$

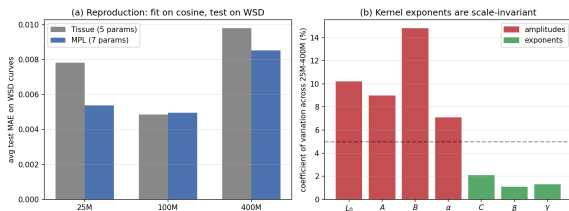
A **Gamma spectrum**  $\propto \lambda^{\beta-1} e^{-\lambda/\lambda_0}$  makes  $\mathcal{K}(y) \propto 1 - (1 + Cy)^{-\beta}$  with  $C = 2\lambda_0$  — MPL's kernel.

**Each parameter now has an identity:**

|                                         |                                                   |
|-----------------------------------------|---------------------------------------------------|
| <b>Exponents</b> $\{\alpha, \beta, C\}$ | = functions of the spectral <i>shape</i>          |
| <b>Amplitudes</b> $\{L_0, A, B\}$       | = set by irreducible loss + spectral <i>scale</i> |

**Prediction.** If the spectrum keeps its shape as model size  $N$  grows, the **exponents are scale-invariant** and only the amplitudes drift with  $N$ .

# Result 1: the kernel exponents are scale-invariant

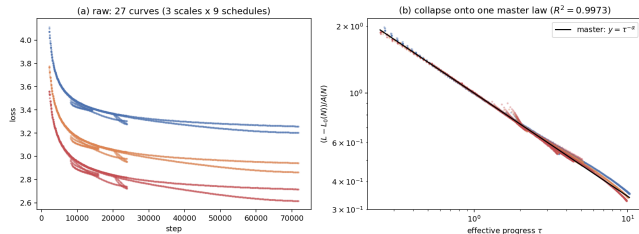


Fitting MPL per scale and comparing parameters across a  $16\times$  size range:

- Exponents  $C, \beta, \gamma$ : CV = 2.1%, 1.1%, 1.3% (essentially constant).
- Amplitudes  $L_0, A, B$ : CV = 9–15% (drift with  $N$ ).
- $\alpha$  (initialization-weighted, predicted least stable of the exponents): 7%.

**The prediction is confirmed:** the shape of the annealing response is a near scale-invariant constant; only the amplitudes carry the size dependence.

## Result 2: all 27 curves collapse onto one master curve



Algebraically MPL is a *single* power law  $L = L_0(N) + A(N) \tau^{-\alpha}$  in the **effective progress**

$$\tau = [S^{-\alpha} + (B/A)LD]^{-1/\alpha},$$

which folds the schedule's annealing into one variable.

Using **one shared exponent set** and only per-scale amplitudes, all **27 curves** (3 scales  $\times$  9 schedules) collapse onto  $y = \tau^{-\alpha}$  with  $R^2 = 0.997$ .

**Analysis.** The collapse succeeds *only because a single exponent set fits every scale* — a direct visual proof of scale invariance. There is essentially **one** loss curve; model size and schedule merely reparameterize the axes.

Code: repro/universal\_collapse.py

## Attempt 1 — SC-MPL: a scale-conditioned law

**Idea.** Since the exponents are scale-invariant, **borrow them from cheap small models** and fit only the 3 amplitudes  $\{L_0, A, B\}$  on the target scale — cutting a new scale from 7 free parameters to 3.

| Scale | MPL            | SC-4    | SC-3    |
|-------|----------------|---------|---------|
| 100M  | <b>0.00435</b> | 0.00451 | 0.00525 |
| 400M  | <b>0.00484</b> | 0.00760 | 0.00786 |

fair comparison, official split, avg test MAE

**Honest result.** Competitive at 100M (wins 4/6 curves with far fewer parameters) but **does not beat a well-fit MPL at 400M**: the residual drift (chiefly in  $\alpha$ ) costs more than the parameter economy gains.

**A methodological caution we discovered.** An earlier protocol that *re-fit* the MPL baseline into a poor optimum made SC-MPL look 20% better; giving the baseline its **true** optimum erases that “win.” *An under-optimized baseline can fabricate a false result.*

## Attempt 2 — Q-MPL: a new discrete-step asymptotic

**New angle.** Tissue and MPL both use the continuum approximation, leaving the single time variable  $S = \sum \eta$ . Keeping the **next discrete-step order** introduces a *second* time variable:

$$\prod (1 - \eta\lambda)^2 \approx e^{-2\lambda S - \lambda^2 Q}, \quad Q = \sum \eta^2.$$

Expanding the bias integral yields a **derived correction**  $\propto Q S^{-(\alpha+2)}$ . We test Q-MPL = MPL +  $E \cdot Q S^{-(\alpha+2)}$  (8 params).

| Scale | MPL            | Q-MPL   | $E^*$       |
|-------|----------------|---------|-------------|
| 25M   | <b>0.00715</b> | 0.00715 | $\approx 0$ |
| 100M  | <b>0.00386</b> | 0.00386 | $\approx 0$ |
| 400M  | <b>0.00669</b> | 0.00824 | overfit     |

**Honest result.** The data drive  $E \rightarrow 0$  at 25/100M (the term is unnecessary), and when free it **over-fits cosine and worsens WSD** at 400M. **Q-MPL does not beat MPL.**

# Verdict: MPL is near-optimal on the current data

Two *principled* extensions — scale-conditioning and a new  $(S, Q)$  asymptotic — both fail to improve a well-fit MPL. The universal collapse ( $R^2 \approx 0.997$ ) shows MPL already captures essentially all reproducible structure.

## Honest conclusion

On these curves MPL is near-optimal: the residual ( $\text{MAE} \sim 0.003$ ) carries no structure that a derived correction exploits.

**Why, and what would change it.** MPL was developed and tuned on exactly this kind of (relatively mild) data. Beating it would likely require data that **exposes MPL's failure modes** — more extreme schedules or more model scales — which is the natural data bottleneck of this study.

# MPL's one empirical parameter: $\gamma$ is essential...

The factor  $\eta_k^{-\gamma}$  is the only part of MPL our derivation does *not* produce. Is it actually needed?

| Scale | full MPL | $\gamma=0$ |
|-------|----------|------------|
| 25M   | 0.0032   | 0.0119     |
| 100M  | 0.0038   | 0.0144     |
| 400M  | 0.0049   | 0.0162     |

real data, test MAE

Holding all exponents fixed and only setting  $\gamma=0$  (then re-fitting the amplitudes) makes the prediction **3–4× worse**.

**So  $\gamma$  is essential** — the original authors were right to include it. But our leading-order derivation predicted  $\gamma=0$ . Where does it come from?

... but it lives at the edge of stability

**Where the LR-magnitude dependence hides.** In  $S$ -units the leading kernel has no  $\eta$  factor ( $\Rightarrow \gamma=0$ ). It appears one order down, in the second time variable  $Q = \sum \eta^2$ . But the term  $\lambda^2 Q$  matters only when  $\eta\lambda \sim 1$  — the highest-curvature modes, i.e. the **edge of stability**, where the linearized SGD recurrence *diverges* rather than relaxes.

**Simulation confirms it.** We simulate SGD on a power-law spectrum and fit MPL vs. a  $\gamma=0$  variant, ablating beyond-quadratic ingredients (noise models, high-curvature weighting, progressive sharpening,  $\lambda \propto \eta^{-s}$ ):

- $\gamma=0$  *always* fits as well as full MPL (e.g. const noise 0.023 vs. 0.019); the fitted  $\gamma$  never tracks the injected scaling.
- Sharpening (the right sign) pushes modes past  $\eta\lambda=1$  and the simulation *diverges* — a numerical signature of the same boundary.

## Result

$\gamma$  is a **non-linear, edge-of-stability effect** outside any spectral quadratic-SGD model — the precise reason it can be fit but not derived.



# Conclusion & takeaways

## What we did.

- Reproduced Tissue & MPL (cosine→WSD), matching published accuracy.
- Derived MPL from an SGD-spectrum model, giving its parameters physical meaning and a falsifiable prediction.

## Main findings.

- **Scale invariance & universality:** the kernel exponents are constant to 1–2%; all 27 curves collapse onto one master power law ( $R^2=0.997$ ) — there is essentially one loss curve.
- **Honest method development:** two principled improvements (SC-MPL, Q-MPL) do not beat a well-fit MPL; MPL is near-optimal on current data. We also show how an unfair baseline fabricates a false win.
- **The limit of the theory:** MPL's one empirical parameter  $\gamma$  is essential yet underivable from quadratic theory — it is an edge-of-stability effect.

**Next steps.** Test the universality across more scales & model families; model the edge-of-stability origin of  $\gamma$ .

# Reproducibility

All results are reproduced by scripts in the repository:

| Result                         | Command                                                    |
|--------------------------------|------------------------------------------------------------|
| Reproduction (Tab. 1, Fig. 1a) | <code>repro/reproduce_cosine_to_wsd.py</code>              |
| Universal collapse (Fig. 2)    | <code>repro/universal_collapse.py</code>                   |
| SC-MPL fair comparison         | <code>repro/validate_theory.py</code>                      |
| Q-MPL new asymptotic           | <code>repro/qmpl.py</code>                                 |
| $\gamma$ & edge of stability   | <code>repro/validate_theory.py, sgd_spectrum_sim.py</code> |
| Full derivation                | <code>docs/core/derivation.md</code>                       |

```
pip install -r requirements.txt; data ships in external/MultiPowerLaw/.
```

Thank you!      Questions welcome.